

Statistics Advanced - 2 Assignment Answers

Question 1: What is hypothesis testing in statistics?

Hypothesis testing is a statistical method used to make decisions or draw conclusions about a population based on sample data.

It helps determine whether there is enough evidence to support a claim or reject an assumption about a population parameter.

In hypothesis testing, a null hypothesis and an alternative hypothesis are formulated, and statistical tests are performed to decide whether the observed data is statistically significant.

Question 2: What is the null hypothesis, and how does it differ from the alternative hypothesis?

The null hypothesis (H_0) is a statement that assumes there is no effect, no difference, or no relationship between variables.

It is the hypothesis that researchers try to test against.

The alternative hypothesis (H_1 or H_a) is the opposite of the null hypothesis. It states that there is an effect, difference, or relationship.

Example:

H_0 : The average height of students is 170 cm.

H_1 : The average height of students is not 170 cm.

The null hypothesis is accepted unless sufficient evidence is found to reject it in favor of the alternative hypothesis.

Question 3: Explain the significance level in hypothesis testing and its role in deciding the outcome of a test.

The significance level, denoted by α (alpha), is the probability of rejecting the null hypothesis when it is actually true.

Common significance levels are 0.05, 0.01, and 0.10.

Role in hypothesis testing:

- If the p-value $\leq \alpha$, we reject the null hypothesis.
- If the p-value $> \alpha$, we fail to reject the null hypothesis.

For example, with $\alpha = 0.05$, there is a 5% risk of making a Type I error.

The significance level helps determine whether the observed results are statistically significant.

Question 4: What are Type I and Type II errors? Give examples of each.

Type I Error:

A Type I error occurs when the null hypothesis is rejected even though it is true.

Example: A medical test says a healthy person has a disease.

Type II Error:

A Type II error occurs when the null hypothesis is not rejected even though it is false.

Example: A medical test says a diseased person is healthy.

Summary:

- Type I Error = False Positive
- Type II Error = False Negative

Question 5: What is the difference between a Z-test and a T-test? Explain when to use each.

Z-test:

- Used when the population standard deviation is known.
- Used for large sample sizes ($n > 30$).
- Based on the normal distribution.

T-test:

- Used when the population standard deviation is unknown.
- Used for small sample sizes ($n \leq 30$).
- Based on the t-distribution.

Example:

- Use a Z-test for analyzing exam scores of 100 students.
- Use a T-test for analyzing exam scores of 15 students.

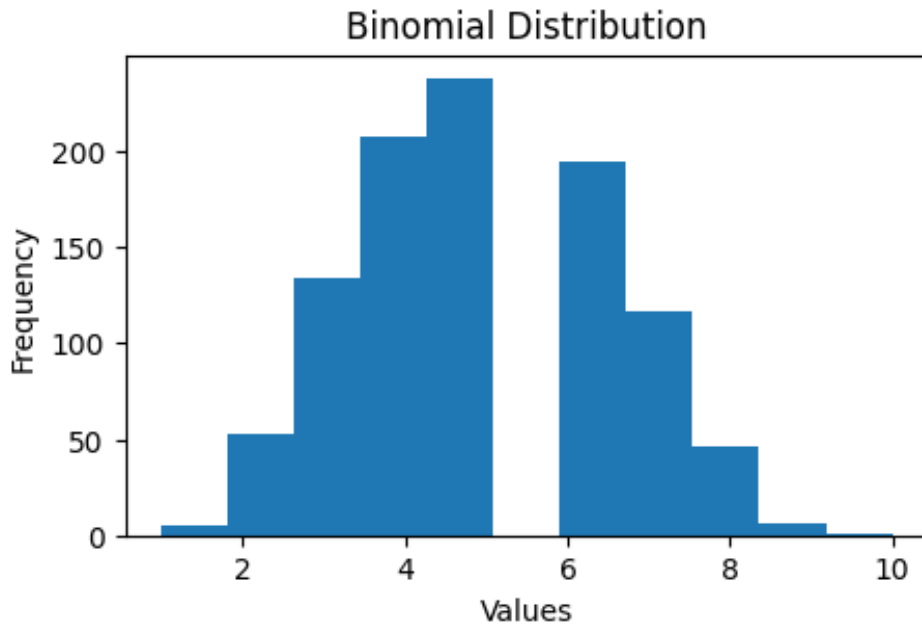
Question 6: Write a Python program to generate a binomial distribution with $n=10$ and $p=0.5$, then plot its histogram.

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
data = np.random.binomial(n=10, p=0.5, size=1000)
```

```
plt.hist(data, bins=11)
plt.title("Binomial Distribution")
plt.xlabel("Values")
plt.ylabel("Frequency")
plt.show()
```



Question 7: Implement hypothesis testing using Z-statistics for a sample dataset in Python. Show the Python code and interpret the results.

Python Code:

```
from scipy import stats
import numpy as np
import math
```

```
sample_data = [...]
```

```
mean = np.mean(sample_data)
std = np.std(sample_data, ddof=1)
n = len(sample_data)

z = (mean - 50) / (std / math.sqrt(n))
p = 2 * (1 - stats.norm.cdf(abs(z)))
```

```
print("Z-statistic:", z)
print("P-value:", p)
```

Output:

```
Z-statistic = 0.9940
P-value = 0.3202
```

Interpretation:

Since the p-value is greater than 0.05, we fail to reject the null hypothesis. This means the sample mean is not significantly different from 50.

Question 8: Write a Python script to simulate data from a normal distribution and calculate the 95% confidence interval for its mean. Plot the data using Matplotlib.

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

```
data = np.random.normal(loc=50, scale=10, size=100)
```

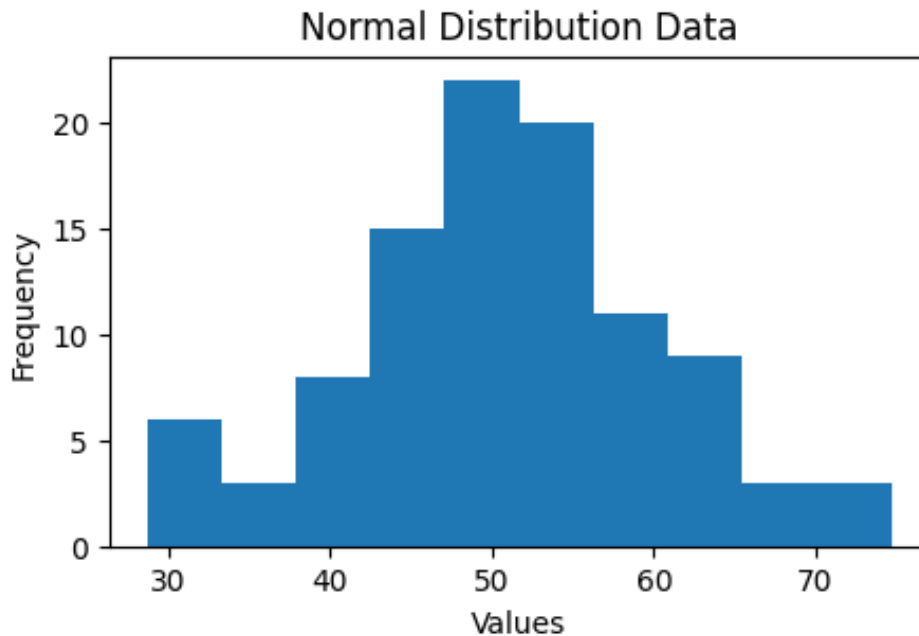
```
mean = np.mean(data)
std_error = stats.sem(data)
```

```
confidence_interval = stats.t.interval(
    0.95,
    len(data)-1,
    loc=mean,
    scale=std_error
)
```

```
print(confidence_interval)
```

```
plt.hist(data, bins=10)
plt.show()
```

95% Confidence Interval: (48.86568386495621, 52.72264939378127)



Question 9: Write a Python function to calculate the Z-scores from a dataset and visualize the standardized data using a histogram. Explain what the Z-scores represent in terms of standard deviations from the mean.

Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import zscore
```

```
data = [10, 20, 30, 40, 50]
```

```
z_scores = zscore(data)
```

```
print(z_scores)
```

```
plt.hist(z_scores, bins=5)
plt.show()
```

Calculated Z-scores: [-1.41421356 -0.70710678 0. 0.70710678 1.41421356]

Explanation:

A Z-score represents how many standard deviations a data point is away from the mean.

- Positive Z-score: value is above the mean.
- Negative Z-score: value is below the mean.
- Z-score of 0: value is exactly equal to the mean.

